Control Variates for Variance Reduction

→ Monte Carlo

↳ powerful for estimating quantities in stochastic models

↳ high variance in estimates, makes us require large no. of simulation, to get accurate results

↳ large no. of simulation

⇒ computation expensive

↳ so use control variates for variance reduction

→ control variate

↳ introducing auxiliary variable X

↳ to improve simulation accuracy

correlated with the quantity of Interest, say Y

↳ X has a known E[X] value.

Adjusted Estimate

Using X, the adjusted estimator for Y is

$$Y_{CV} = Y - c(X - E[X]), \quad c = \frac{Cov(Y,X)}{Var(X)}$$

$$\because E[Y_{CV}] = E[Y] - E[X - E[X]]$$

$$= E[Y] - (E[X] - E[X])$$

$$\boxed{E[Y_{CV}] = E[Y]}$$

and

$$Var(Y_{CV}) = Var(Y) + c^2 Var(X - E[X]) - 2c\, Cov(Y, X - E[X])$$

$$= Var(Y) + c^2 Var(X) - 2c\, Cov(X,Y)$$

$$= Var(Y) + \frac{(Cov(Y,X))^2}{Var(X)} - 2\frac{(Cov(X,Y))^2}{Var(X)}$$

$$= Var(Y) - \left(\frac{Cov(X,Y)}{Var(X)}\right)^2 Var(X)$$

$$\boxed{Var(Y_{CV}) = Var(Y) - c^2 Var(X)}$$

→ Same Expected estimate, with lower variance
↳ so more precise estimation

Example:

European Option Pricing (GBM)

Geometric Brownian Motion

→ payoff at maturity T, for a european call option

$$C = \max(S_T - K, 0) * \underbrace{\exp(-rT)}_{\text{discounted back}}$$

S_t - Underlying price at t

K - strike

T - time to maturity

GBM dynamics of S_t

$$S_t = S_0 \exp\{(\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\}$$

μ - drift of underlying stock $= r$

σ^2 - volatility of " "

S_0 - initial price of " "

W_t - standard Brownian Motion

↳ $\sqrt{t}\, Z$

$Z \sim N(0,1)$

$E[S_t] = S_0 e^{\mu t}$

$Var(S) = \sigma^2$ ⇒ $S_{t+\Delta t}$

$$= S_t * \exp\{(-\tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z\}$$

Control Variate

↳ Geometric Average of Simulated prices

→ since geometric avg (G) is highly correlated with terminal stock price, it soaks up a lot of the simulation noise

$$X = \left(\prod_{i=1}^{T} S_{t_i}\right)^{1/T}, \quad T\text{- steps of GBM}$$

$$\log X = \frac{1}{T}\sum_{i=1}^{T} \log(S_{t_i})$$

$$\Rightarrow X = \exp\left(\frac{\text{log-sum}}{\text{no-of steps}}\right)$$

→ for a continuous average over $[0,T]$,

$$E[X] = S_0 \exp\left\{\frac{1}{2}\left(\mu - \frac{\sigma^2}{6}\right)T\right\}$$

→ now, Y - payoff and X - Geometric Average

$$Y_{CV} = Y - \underbrace{c(X - E[X])}_{\text{correction term that reduces variance}}$$

run MC simulations and take $E[Y_{CV}]$ as the estimate